Vishmith Suranjaya

Computer Science Undergraduate

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Summary

Motivated Computer Science undergraduate with a strong passion for software development and creating meaningful digital experiences that blend creativity with functionality. Skilled in React.js and Python, with a keen interest in continuous learning and problem-solving. Known for a thoughtful, detail-oriented approach and driven by the belief that growth begins beyond comfort zones.

Education

Uva Wellassa University, BSc(hons) in Computer Science and Technology — Expected September 2026

- CGPA: 3.77/4.0
- **Related Coursework:** Data Structures and Algorithms, Object Oriented Programming, Software Engineering, Software Architecture Design Patterns, Web Application Development, Operating Systems

Karandeniya Central College, Advanced Level (A/L)

- Physical Science Stream

Skills

React.js, Tailwind CSS, JavaScript, C

Python, Django, Java, Maven, PHP

MySQL, PostgreSQL, Firebase

Flutter, Kotlin

Git, Github, Docker

Jupyter Notebook

Research Project

Energy-Efficient Deep Learning on ARM Architectures using Spiking Neural Networks (SNNs) — 2025 – Present

Technologies Using: Deep Learning, Nvidia Jetson Nano, Python

Uva Wellassa University

- Investigating the feasibility of deploying optimized Spiking Neural Networks (SNNs) on ARM-based embedded systems for energy-efficient AI inference.
- Developing a baseline Artificial Neural Network(ANN) and SNN using the N-MNIST dataset and applying model-level optimizations such as quantization and pruning to improve latency, memory, and power efficiency.
- Conducting performance benchmarking to evaluate accuracy, energy consumption, and computational cost against conventional deep learning models.
- Exploring hardware–software optimization strategies to leverage ARM features for efficient SNN deployment.
- Assessing scalability and potential integration of SNN-based models for real-time edge AI applications in IoT, robotics, and autonomous systems.

Projects

Infant Guardian - Edge AI healthcare system for real-time infant cry classification 01/2026 – Present
Technologies Used: Kotlin, firebase View Project

- Building a lightweight CNN Model
- Developing a cross-platform mobile application using Kotlin

Seagrass Srilanka - AI powered Seagrass Conservation Platform 05/2025 – 09/2025
Technologies Used: React, Django, Flask, PostgreSQL, Python, Azure OpenAI View Project

- Built a React frontend integrated with a Django + PostgreSQL backend for dynamic content management.
- Developed a Flask microservice hosting a model for automatic seagrass species recognition.
- Designed an AI-powered digital platform supporting seagrass conservation in Sri Lanka, integrating species identification, educational blogs, and an immersive 3D virtual underwater experience.
- Leveraged the Azure OpenAI API to optimize blog content and enable AI-driven writing assistance for enhanced educational outreach.

Cloudify - Smart Weather Recommendation App 02/2025 – 03/2025
Technologies Used: Flutter, OpenWheatherAPI, Lottie View Project

- Real-Time Weather Data: Fetch and display accurate weather information including temperature, conditions, and city name, with automatic real-time updates.
- Clothing Recommendations: Provide personalized clothing suggestions based on current weather conditions, offering a unique feature that differentiates the app from standard weather platforms.

estateEase - AN Innovative Real-Estate Platform 04/2024 – 07/2024
Technologies Used: PHP, MySQL, JavaScript, HTML, CSS View Project

- An innovative real estate platform that streamlines property transactions with features like bidding, Google Maps integration, advanced search and filter options, secure payments, and an installment calculator.

Soft Skills

Communication Adaptability Problem Solving Teamwork

Certifications

AI/ML Engineer - Stage 1 (*SLIIT*) **Machine Learning Using Python** (*Simplilearn*) **PMP Basics** (*Simplilearn*)

Achievements

Dean's List *Faculty of Applied Sciences, Uva Wellassa University*

Dean's Letter of Appreciation *UWU Expo Exhibition*

- Received a formal letter of appreciation from the Dean for outstanding leadership and contribution.
- Led a 10 member team to implement and showcase an immersive undersea Virtual Reality (VR) experience.

References

Dr. K. P. P. S. PATHIRANA *Senior Lecturer, Department of Computer Science and Informatics, Faculty of Applied Sciences, Uva Wellassa University, Badulla, Sri Lanka.* suneth@uwu.ac.lk
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